

Bass-DDSP v2: label-conditioned transient-assisted hybrid synthesizer

T = control frames, block_size = 256 audio samples, N = T * block_size. Active controls are causal; note_progress, string_id, fret_id, and centroid are excluded.

Data and controls

IDMT-SMT-BASS / riff generator
trim -> crossfade -> aligned labels

frame extraction

Frame controls, all at (B,T,*)

- f0(t): label Hz, (B,T,1)
- loudness(t): z-score, (B,T,1)
- articulation_id(t): observed class, (B,T)
- gate(t): active note mask
- onset_strength(t): HPSS [0,1]
- offset(t): note-end pulse
- note_age(t): causal seconds since onset
- periodicity(t): harmonicity prior/conf.

Target waveform
x_target: (B,N,1)

Frame-rate encoder / decoder

f0 Hz -> MIDI [0,1]
Pitch MLP -> 64 ch

Loudness z-score
Loudness MLP -> 64 ch

ArticulationEncoder
embedding(24) + onset/offset/gate/
note_age/periodicity -> z(t), 64 ch

z(t) MLP -> 256 ch

concat features
64 + 64 + 256 = 384

causal sequence model
GRU H=256
(Mamba option)

Out MLP
[GRU, f0, loudness, z]
-> hidden h(t): (B,T,256)

Audio-rate differentiable synthesis

audio-rate
control expansion
upsample/interp
note_age + sample offset

branch modulators
loudness_gain
harmonic_gate
transient_window
fixed branch_gain

Sustain branch: DWTS-style wavetable
h(t) -> 16-table attention + amplitude
f0_audio -> phase accumulator
512-sample learned wavetable lookup

sustain *= gate * fade_in(age)
* loudness_gain * harmonic_gate

Noise branch: subtractive noise
h(t) -> 65 band magnitudes -> IR
white noise -> FFT convolution

noise *= gate * fixed branch_gain
continuous residual noise path

Transient branch: DCT-bank prototype
articulation_id -> class DCT coeff row
IDCT -> 300 ms style waveform; h(t) -> gain

transient *= gate * fixed window(note_age)
* onset-strength velocity

Output and losses

Training loss
vs x_target
multi-scale STFT
+ log-frame-RMS

sum
sustain
+ noise
+ transient

optional reverb
OFF by default

Reconstruction
x_hat: (B,N,1)

Diagnostics
branch RMS, fixed gains,
attention, gates, W&B

